

CS253 - Data Structures and Algorithms for DS/AI

Spring 2025

PSO 1

Question 1

Let c be the cost of calling the function `WORK`. That is, the cost of the function is constant, regardless of the input value. Determine the respective closed-form $T(n)$ for the cost of calling `WORK`.

```
1: function A1( $n : \mathbb{Z}^+$ )
2:    $val \leftarrow 0$ 
3:   for  $i$  from 1 to  $n$  by multiplying by 3 do
4:     for  $j$  from  $i$  to  $i^2$  do
5:        $val \leftarrow val + \text{WORK}(n)$ 
6:     end for
7:   end for
8:   return  $val$ 
9: end function
```

Question 2

Derive the closed-form $T(n)$ for the value returned by the following algorithm:

```
1: function A2( $n : \mathbb{Z}^+$ )
2:    $sum \leftarrow 0$ 
3:   for  $i$  from 0 to  $n^4 - 1$  do
4:     for  $j$  from  $i$  to  $n^3 - 1$  do
5:        $sum \leftarrow sum + 1$ 
6:     end for
7:   end for
8:   return  $sum$ 
9: end function
```

Question 3

(a) The following statements are true or false?

1. $n^2 = O(5^{\log n})$
2. $\frac{\log n}{\log \log n} = O(\sqrt{\log n})$
3. $n^{\log n} = \Omega(n!)$

(b) Sort the following functions in increasing order of asymptotic (big-O) complexity:

$$f_1(n) = n^{\sqrt{n}}, \quad f_2(n) = 2^n, \quad f_3(n) = n^{10} \cdot 2^{n/2}, \quad f_4(n) = \binom{n}{2}$$

Question 4

- (a) Show that $\max\{f(n), g(n)\} = \Theta(f(n) + g(n))$ for any $f(n)$ and $g(n)$ that eventually become and stay positive.
- (b) Give an example of f and g such that f is not $O(g)$ and g is also not $O(f)$.